# GEORGE DATSERIS – CURRICULUM VITAE

Senior Lecturer in the University of Exeter,
Faculty of Science, Environment and Economy,
Department of Mathematics and Statistics
g.datseris@exeter.ac.uk, https://datseris.github.io/

## ACADEMIC APPOINTMENTS

| | |
|---|---|
| 10/2025 - ongoing | **Senior Lecturer,** *Department of Mathematics and Statistics, University of Exeter* |
| 10/2023 - 09/2025 | **Marie Sklodowska Curie Postdoctoral Fellow**, *Department of Mathematics and Statistics, University of Exeter, mentors: Peter Ashwin, Geoffrey Vallis* |
| 01/2023 - 09/2023 | **Royal Society Newton International Fellow,** *Department of Mathematics and Statistics, University of Exeter, mentors: Peter Ashwin, Geoffrey Vallis* |
| 01/2020 - 12/2022 | **Postdoc,** *Atmosphere in the Earth System Department (dir Bjorn Stevens), Max Planck Institute for Meteorology, supervisor: Bjorn Stevens* |
| 10/2019 - 12/2019 | **Postdoc,** *Living Matter Physics Department (dir Ramin Golestanian), Max Planck Institute for Dynamics and Self Organization, supervisor: Ragnar Fleischmann* |
| 04/2016 - 09/2019 | **PhD in Physics: "ballistic electron transport in graphene nanodevices and billiards",** *Nonlinear Dynamics Department (dir Theo Geisel), Max Planck Institute for Dynamics and Self-Organization and University of Goettingen, supervisors: Theo Geisel, Ragnar Fleischmann* |
| 10/2015 - 04/2016 | **Research Assistant,** *Nonlinear Dynamics Department (dir Theo Geisel), Max Planck Institute for Dynamics and Self-Organization* |
| 2009 - 2016 | **BSc and MSc in Physics, majored in Condensed Matter,** *Physics Department, National and Kapodistrian University of Athens* |

## BRIEF SUMMARY

I am a creative, self-driven independent researcher, with a unique skill set that combines the mathematics of nonlinear dynamics and the physics of complex systems with numerical modelling, spatio/temporal data analysis, and climate science.

During my PhD I worked on classical and quantum electron transport in graphene nanodevices, but also nonlinear timeseries analysis and music fluctuations and their human perception. After my PhD, I wanted to move my research focus into climate science. To this end I joined the Max Planck Institute for Meteorology as a Postdoc. This substantial career change was challenging, but I was able to bring valuable new perspectives from nonlinear dynamics and complex systems into climate science, primarily the study of large-scale cloudiness and its interaction with the energy balance of the Earth system. At the end of my postdoc, I wanted to carry on working in clouds & climate but also connect more with my roots in nonlinear dynamics and physics. I won two fellowships to support this: the Royal Society International Newton Fellowship and the Marie Sklodowska Curie Postdoctoral Fellowship. They brought me to the University of Exeter, where I combined my passion for developing conceptual understanding of the physical world with understanding the interplay between clouds and climate. Most of my research is split between nonlinear dynamics and cloud-climate interactions. On one front, I am developing a completely new perspective for understanding the response of dynamical systems to finite perturbations and to changes in their parameters. On the other front, I combine conceptual climate models, satellite and in-situ data, and large eddy simulations, with the goal of understanding how clouds respond to change or generate themselves change in the climate system. My current research focus is the treating clouds as systems capable of undergoing critical transitions, and what this means for the climate system and other so-called tipping elements.

# RESEARCH

## AREAS OF RESEARCH

I have a broad interest in nonlinear dynamics and complex systems and their application to understand the physical world, primarily the climate system, at a conceptual level. I value interdisciplinarity and diversity in research and science. My current main research areas are:

1. **Clouds and climate: conceptual modelling, sensitivity, and critical transitions.** Clouds are both beautiful and one of the most important components of the Earth's climate. My research focuses on understanding how clouds change and generate change in the climate system. To do this I model clouds as multistable systems capable of undergoing critical transitions on a wider scale. I incorporate analysis of low-dimensional nonlinear models, large eddy simulations, and observational data. My main goal is to identify under what circumstances clouds undergo critical transitions, what is the impact of this on the climate system, and whether we can forecast such events.
2. **Nonlinear dynamics: multistability, continuation, multi-parameter spaces.** Nonlinear dynamical systems are used to model everything, from power grids to clouds. My work focuses on multistable models, where alternative states can co-exist, and critical transitions can be generated across such states. My research revolves around how to study multistability as system parameters vary. To this end I have developed the groundbreaking technique of global continuation, a way to continue information from the whole state space across an arbitrary parameter curve. I am now focused on scaling these methodologies to multi-parameter spaces to provide tools for studying realistic complex systems that often depend on 10++ parameters.
3. **Timeseries analysis: from linear to associative.** I've applied timeseries analysis on a variety of real world data, from satellite radiance measurements to stock market prices and timeseries of live music performances. I've accumulated a large toolset for timeseries analysis: linear (spectral filters, autocorrelations), power-law (Levy walks / DFA), autoregressive (ARMAs), decomposing (PCA/DMC), nonlinear (delay embedding, fractal), complexity-based (sample, permutation entropies), surrogates hypothesis testing, geospatial (variograms), change point detection / early warning signals, and associative (causal). My latest timeseries project is applying associative timeseries analysis to create a graph of "causal" cloud controlling factors for understanding the coupling of cloudiness to the broader environment.

## RESEARCH GRANTS, INTERNATIONAL

2023 **Marie Sklodowska Curie Postdoctoral Fellowship**, (€220,908.48)
*awarded by the European Commission, and funded by the UK Research and Innovation*

2022 **Walter Benjamin Fellowship**, (€74,004.72), *awarded by the German Research Foundation (DFG), declined*

2022 **Royal Society Newton International Fellowship** (£131,250), *awarded by the Royal Society*

2018 **IMPRS Travel Grants,** (€1500), *for travelling to JuliaCon2018 (UK) and deRSE19 (Germany)*

2016 **International Max Planck Research School (IMPRS) Excellence Fellowship** (circa €1,800/month)
*awarded by the Physics of Biological and Complex Systems PhD School, funded my PhD*

## PUBLICATIONS

### BOOKS AND BOOK CHAPTERS

2026 **Contributed chapter "Nonlinear dynamics and climate: multistability and tipping",** *contributed to "Handbook of Nonlinear Dynamics", WorldScientific, Edited by Norbert Hoffman, in production*

2022 **Nonlinear Dynamics: a concise introduction interlaced with code,** *Datseris & Parlitz, Springer-Nature, Undergraduate Lecture Notes in Physics Series; postgraduate or undergraduate level textbook, covering a semester-long course in nonlinear dynamics; includes 250+ exercises, 100+ figures, 10+ GUI apps, 100+ polling-style questions for promoting student engagement, 20+ code snippets for Julia code. Textbook has approximately 60,000 digital download requests from SpringerLink.*

### REVIEW ARTICLES

2023 **Estimating fractal dimensions: A comparative review and open source implementations,** *Datseris, Kottlarz, Braun, Parlitz, in Chaos, doi: 10.1063/5.0160394*

### RESEARCH ARTICLES

For a manually curated list of my published articles and basic bibliometrics visit my Google Scholar page:

https://scholar.google.com/citations?user=5U_IlXcAAAAJ&hl=en

## RESEARCH IMPACT AND KNOWLEDGE EXCHANGE

### SOFTWARE

I have lead-developed over 20 individual software packages. The list below outlines the ones that are largest in amount of content and user bases (date listed corresponds to first commit):

2017 **DynamicalSystems.jl,** *a general purpose software library for the entire field of nonlinear dynamics and nonlinear timeseries analysis; has by far the largest number of diverse algorithm implementations and most open-source individual contributors (~25); has 900 GitHub stars and an estimate of 25,000 users; the software is cited 145 times*

2019 **Agents.jl,** *General purpose software library for agent based modelling. Proven via objective measures in peer-reviewed article (Datseris, Vahdati, DuBuis, Simulation, 2024, 10.1177/00375497211068820) and automated comparison exposed in the documentation, to be the most performant, most featureful, and simplest, general purpose open source software for agent based modelling. Has an estimate of 19,500 users, 850 GitHub stars. The software is cited 115 times.*

2019 **DrWatson,** *Julia-based software for managing scientific projects, making them accessible, reproducible, and easy to use and navigate. Focused primarily on computational sciences but can be used to help any project; has 40,000 downloads (upper user estimate), one of the most popular packages for the Julia language; uniqueness of the software: useful features directly relevant for day-to-day project management while being minimally invasive and not requiring workflow alterations*

2023 **ComplexityMeasures.jl,** *software for* entropic and complex timeseries analysis, formally a component of DynamicalSystems.jl but with its own publication; *published in PLOS One (2025, 10.1371/journal.pone.0324431), the associated peer reviewed article shows ComplexityMeasures.jl to have 63x more implemented measures and up to 2800x higher computational performance from the second-best option (an already established software "EntropyHub" with ~60 citations)*

Beyond these, other notable software, all of which have been used in published research articles, are:

2017 **DynamicalBilliards.jl,** *an easy-to-use, modular, extendable and absurdly fast Julia package for dynamical billiards in 2D*

2016 **MIDI.jl,** *a Julia library for handling MIDI files*

2018 **DelayEmbeddings.jl,** *library for performing and optimizing delay coordinate embeddings*

2022 **Attractors.jl,** *find attractors (and their basins) of dynamical systems. Perform global continuation. Study nonlocal stability*

2018 **MusicManipulations.jl,** *Manipulate, humanize, quantize and analyze music data*

2019 **TimeseriesPrediction.jl,** *forecast temporal and spatiotemporal timeseries using local reconstructed states*

2018 **TimeseriesSurrogates.jl,** *a Julia package for generating timeseries surrogates*

2020 **ClimateBase.jl,** *tools to analyze and manipulate climate (spatiotemporal) data*

### KNOWLEDGE EXCHANGE

2025 **Consultancy work for DH-BG Systematic Trading LTD**
*funding 20 hours of my time and 3 months part time work for MSc student for paid consultancy, resulted from a Knowledge Exchange initiative established by the KEHub*

2025 **Consultancy workshop for NFDI4Earth consortium (www.nfdi4earth.de), hired by GFZ Helmholtz-Zentrum fur Geoforschung,** *hired to perform 3 out of 6 sections of my Good Scientific Code Workshop*

2019 - ongoing **Google Summer of Code Internships,** *funded by Google to supervise students over summer; students contribute mathematical algorithms to open source libraries under my supervision (see undergraduate research below)*

## EDUCATION

### TAUGHT COURSES/MODULES

2025 **Computational Nonlinear Dynamics, Term 1, Year 3 (MTH3039)** *module lead and solo lecturer; 33 lecture hours; 22 office hours*

2024 **Computational Nonlinear Dynamics, Term 1, Year 3 (MTH3039)**
*lecturer for second half of the module, led by Jan Sieber; 16 lecture hours; 11 office hours*

2023 **Mathematics Group Projects, Term 2/3, Year 3 (MTH3035)**
*led one of the groups with topic "complexity timeseries analysis for stock market timeseries" for the MTH3035 module led by James Screen; 16 lecture hours*

2023 **Introduction to Juia, University of Exeter,**
*given as part of the "Coding for Reproducible Research" training programme of University of Exeter, 12 lecture hours*

2016 + 2017 **Tutoring for "Introduction to Complex Systems", University of Goettingen**
*3rd year BSc course in Physics faculty, module-led by Ulrich Parlitz; performed twice*

2014 **Chaos in 1D and 2D maps, National and Kapodistrian University of Athens,**
*9 hour guest lecture on "Chaos in 1D and 2D Maps" for 4th year undergrads*

### NEW DEVELOPED COURSES AND INNOVATIVE PRACTICES IN EDUCATION

2025 **Fellowship of the Higher Education Academy award (APA),**
*UK-specific qualification for academic practices, primarily teaching-focused*

2025 **Fully redesigned MTH3039 "Computational Nonlinear Dynamics"**
*Content of module MTH3039 of University of Exeter was reworked brand new from the ground up, adding significantly more quantity of more diverse content, without sacrificing student learning, as evidenced by constancy in student grades. The module is now based on the Julia programming language (taught during module time), a free and superior alternative to MATLAB or Python for computational nonlinear dynamics. The assessment format was reworked: submitted coursework solutions format is the Jupyter Notebook (which I also taught basic usage of during the module. Finally the module now teaches students how to use the internationally acclaimed software for nonlinear dynamics "DynamicalSystems.jl".*

2022 - ongoing **Usage of digital resources for promoting student engagement,**
*in modules I lead I utilize two advanced digital resources to enhance student engagement: I create interactive Graphical User Interfaces (GUIs) that students can run on their computers and explore taught concepts; I create online interactive polling/questioning resources where students anonymously contribute; these resources provably promote student engagement: 75+% of the students of my modules participate*

2024 **EDI talk "how to make accessible presentations for the visually impaired"**
*invited to present this to the programme meetings of Mathematics department of the University of Exeter by the director of Education and Student Engagement, Layal Hakim*

## WORKSHOPS & OTHER EDUCATIONAL RESOURCES

| | |
|---|---|
| 2024 | **Writing and maintaining an exceptional Documentation,** *3-hour workshop presented at JuliaCon 2024, also available on YouTube* |
| 2023 (x2) | **DynamicalSystems.jl and Attractors.jl- hands-on workshop on a new paradigm of dynamical systems stability,** *full-day workshop on using the aforementioned software; first given in Max Planck Institute for Evolutionary Biology (Ploen, Germany); then repeated in "Non-autonomous Dynamics in Complex Systems: Theory and Applications to Critical Transitions" workshop (Dresden, Germany)* |
| 2021 | **Educational Mathematics Video as part of SoME#1 (online),** *My contribution "Explanation of the butterfly effect and deterministic chaos using billiards" was ranked 20th out of the 777 submissions and accumulated 28,000 views on YouTube* |
| 2020 - ongoing | **Julia Zero2Hero Workshop,** *an open-access, internationally acclaimed three-day intensive workshop serving as an introduction to the Julia Programming language. Has been performed numerous times in MPIMET, Exeter, Ploen, and online. It is a constantly evolving, constantly improving educational resource. Available on YouTube with ~25,000 views. It includes ~20 exercises for participants to hone their skills.* |
| 2019 - ongoing | **Good Scientific Code Workshop,** *an open access six-week block-based workshop for creating scientific code bases that are Clear, Easy to understand, Well-documented, Reproducible, Testable, Reliable, Reusable, Extendable, and Generic. Workshop GitHub page has 300 stars, YouTube recordings of workshop have ≈ 6,000 views. It is a living, constantly evolving educational resource. Latest iteration: Invited by GFZ Helmholtz-Zentrum fur Geoforschung to give a workshop on 3 out of the 6 sessions to the NFDI4Earth consortium (www.nfdi4earth.de)* |
| 2017 + 2019 | **Introduction to DynamicalSystems.jl,** *2-hour long online workshop (livestreamed and uploaded on the Julia Language official YouTube channel) educating on using the DynamicalSystems.jl library for academic or industrial use, performed twice.* |
| 2014 - 2017 (x4) | **Spray Painting Workshop(s),** *12 weekly lectures on introducing the art of spray painting to beginners (with or without previous painting experience with other media); course developed and taught entirely by myself; performed twice at spaces provided by the University of Athens (with permission), and twice at spaces provided by the Max Planck Institute for Dynamics and Self-Organization (with permission)* |
| 2011 - 2014 | **Private Tutoring,** *1-on-1 high school mathematics, physics and chemistry in Greece* |

## EXAMINING/ASSESSING WORK

| | |
|---|---|
| 2026 | **Assessor for MSc to PhD upgrade of Ali Erwy,** *supervised by Jan Sieber* |
| 2025 | **Assessor for MSc to PhD upgrade of Tomas Buckley,** *co-supervised by Peter Ashwin and Paul Ritchie* |

## SUPERVISION (GRADUATE AND POSTGRADUATE RESEARCH)

### POSTGRADUATE RESEARCH (PHD AND POSTDOC)

| | |
|---|---|
| 2025 | **Postdoc: Andreas Morr, co-supervision at 50% time, Technical University Munich,** *Lead supervisor Christian Kuehn; co-supervised on the research project "Computing Resilience Measures in Dynamical Systems"* |
| 2024 | **PhD: Ignacio Del Amo, co-supervision at 20% time, University of Exeter,** *Lead supervisor Mark Holland; successful PhD completion (07/2025); currently postdoc at University of Copenhagen* |
| 2024 | **PhD: Muhammed Fadera, co-supervision at 20% time, University of Exeter,** *Lead supervisor Peter Ashwin; successful PhD completion on 09/2025; currently postdoc at University of Exeter* |
| 2024 | **PhD: Maria Alejandra Ramirez, co-supervision at 20% time, Max Planck Institute for Evolutionary Biology,** *Lead supervisor Andre Traulsen; co-supervised on the research project "Chaos and noise in evolutionary game dynamics"; successful PhD completion on 11/2025* |

### GRADUATE RESEARCH (BACHELORS, MASTERS, INTERNSHIPS)

2025 **MMath individual project: Aryan Paten, solo supervision, University of Exeter**
*Project grade: 60/100; successful MMath completion (11/2025)*

2025 **EPSRC Summer Internship: James Bluck, solo supervision, University of Exeter**
*Successful EPSRC Internship completion (10/2025)*

2025 **EPSRC Summer Internship: Alex Fisher, solo supervision, University of Exeter**
*Successful EPSRC Internship completion (10/2025)*

2025 **GSoC (Google Summer of Code): Georgio Bertone, co-supervision at 50% time, remote**
**GSoC** *are summer internships funded by Google for students to work on open source projects during the north-hemispheric summer time. Co-supervisor Adriano Meligrana; successful GSoC completion (08/2025)*

2024 **GSoC: Jonas Koziorek, solo supervisor, remote**
*successful GSoC completion (08/2024); currently MSc student at MPIMET*

2023 **GSoC: Stefan Vayl, solo supervisor, remote**
*successful GSoC completion (08/2023)*

2021 **GSoC: Aayush Sabharwal, co-supervised at 50% time, remote**
*Co-supervisor Tim DuBois; successful GSoC completion (09/2021);*
*currently senior software engineer at JuliaHub*

2021 **GSoC: V. M. Vasi, co-supervised at 20% time, remote**
*Co-supervisor Avik Sengupta; successful GSoC completion (09/2021)*

2021 **MSc thesis: Isaac Baffour, co-supervised at 50% time, Max Planck Institute for Meteorology**
*Co-supervisor Hauke Schmidt; successful MSc completion (2022); currently data analyst at DKRZ*

2018 **BSc thesis: Lennart Jahn, co-supervised at 80% time, Max Planck Institute for Dynamics & Self-Organization,** *Co-supervisor Theo Geisel; successful BSc completion (2018)*

2018 **BSc thesis: Jonas Isensee, co-supervised at 50% time, Max Planck Institute for Dynamics & Self-Organization,** *Co-supervisor Ulrich Parlitz; successful BSc completion (2018); currently finished a PhD at MPIDS*

2018 **BSc thesis: Lukas Hupe, co-supervised at 50% time, Max Planck Institute for Dynamics & Self-Organization,** *Co-supervisor Ragnar Fleischmann; successful BSc completion (2018); currently finished a PhD at MPIDS*

## LEADERSHIP AND MANAGEMENT

### SOFTWARE LEADERSHIP

2017 - ongoing **Lead developer of JuliaDynamics**
*JuliaDynamics is a GitHub software organization for dynamical and complex systems software written in Julia. Started in 2017, it now has ~60 repositories (software), ~40 members (regular contributors), ~100 one-off contributors, and 100,000+ total software downloads (users upper estimate). As lead dev I coordinate & admin org infrastructure, organize monthly online meetings, scout and invite new members and software, coordinate software interdependencies, curate documentations, help existing members with their software development, and review pull request (code mentorship).*

2024 - ongoing **Organizer of JuliaDynamics monthly meetings**
*I organize the monthly international (online) meetings for "nonlinear dynamics and complex systems with Julia" (Started October 2024, continuing monthly since). These showcase recent research work on the subject by invited speakers and allow for open-ended discussions on software and science. For example, the speaker of November 2025 meeting was Prof. Sergio Roberto Lopes of Universidade Federal do Paraná.*

2018 - 2021 **Lead developer of JuliaMusic,** *similar duties to lead dev. of JuliaDynamics but smaller in scope.*

2018 - ongoing **Broader software community assistance**
*Partly as lead dev of JuliaDynamics, partly as passionate supporter and promoter of open source code, I regularly assist the broader community with coding. I typically review 5 pull requests per month and answer 5-10 user questions per month on the official Julia discussion forums or on GitHub.*

### ACADEMIC LEADERSHIP AND MANAGEMENT

2026 **Minisymposium "Nonlinear and Complex Systems", JuliaCon2026**
*organized solely by myself, composed of 11 invited speakers*

2025 – ongoing **Dynamics Holiday Potluck**
*organizing and hosting community holiday event for the Centre for Systems Dynamics and Control*

2025 **Minisymposium "Multistability and Nonlocal Stability Analysis", Dynamics Days 2025**
*Co-organized with Kalel Rossi, composed of 8 invited speakers, this minisymposium was one of the most successful sessions in the conference Dynamics Days 2025 (measured by number of participating audience)*

2025 - ongoing **Open Research Champion for Mathematics and Statistics, University of Exeter,**
*in this role I disseminate information from UKRN; primarily, I create, organize, and promote activities with the goal of establishing open research, educating on how to better achieve open research, and identifying factors leading to lack of open research*

2024 - ongoing **Dynamics Morning Coffee**
*organizing ongoing weekly informal meetings for the Centre for Systems Dynamics and Control*

2024 **Minisymposium "Multistability and Global Stability Analysis", Dynamics Days 2024**
*Co-organized with Kalel Rossi, composed of 8 invited speakers, this minisymposium was one of the most successful sessions in the conference Dynamics Days 2024 (measured by number of participating audience)*

2023 - ongoing **Hactoberfest at University of Exeter**
*organized and hosted a Hacktoberfest event promoting open source code at the University of Exeter at years 2023 and 2025*

2021 **Panel Discussion Member on Open Science, Lindau 70th Meeting,**
*I was one of the four panelists of the "Open Science" Panel Discussion which took place during the 70th Lindau Nobel meeting. I co-hosted the panel with Dr. Jex and Nobel Laureates Prof. Blackburn and Prof. Schekman. It is available publicly, https://www.mediatheque.lindau-nobel.org/recordings/ 39149/open-science.*

2019 **Hacktoberfest at Max Planck Institute for Dynamics and Self-Organization**

2019 **Goettingen GGNB PhD School Debate Club,** *main organizer*

2018 **Bi-Annual retreat for the IMPRS PhD School for the Physics of Complex Systems,** *co-organizer*

## EXTERNAL RECOGNITION

### HONOURS AND AWARDS

2026 **Editor's Pick in CHAOS**
*Our paper "Computing Resilience Measures in Dynamical Systems" (Morr, Kuehl, Datseris, 2026) was selected as an Editor's Pick, which signifies that the editors felt that our article was above average for the journal*

2024 **Textbook public praise**
*Our textbook "Nonlinear Dynamics" (Datseris and Parlitz 2022) was publicly advertised as best-selling in its discipline on the social platform LinkedIn by the (ex.) executive editor in Physics for Springer, Angela Lahee*

2023 **Cover Article & Featured Article in CHAOS**
*Our paper "Estimating fractal dimensions" (Datseris et al. 2023) was selected as a Featured Article, which signifies that the editors felt that our article was one of the journal's best. The same article then became the cover article for the volume it was published in*

2023 **Featured Article in CHAOS and Showcase in AIP Publishing**
*Our paper "Framework for global stability analysis of dynamical systems" (Datseris, Rossi, Wagemakers 2023) was selected as a Featured Article, which signifies that the editors felt that our article was one of the journal's best. The same article was showcased in the American Institute of Physics (AIP) Publishing Showcase on Kudos*

2022 **Max Planck Institute for Meteorology Distinction**
*awarded for my "exceptional performance in intellectual culture and community building". This award also came with a premium payout (€6,000)*

2022 **Featured Article in CHAOS**
*Our paper Effortless estimation of basins of attraction (Datseris & Wagemakers 2022) was selected as a Featured Article, which signifies that the editors felt that our article was one of the journal's best*

2021 **Editor's Highlight in Eos.org**
*Our paper "Earth's albedo and its symmetry" (Datseris & Stevens 2021, AGU Advances) was selected as an Editor's Highlight in Eos.org (fewer than 2% of AGU journal articles are featured in this way)*

2020 **70th Lindau Nobel Laureate Meeting**
*I was one of the few young scientists selected to participate in the meeting, which provided direct interaction with several Nobel Laureates*

2019 **Summa Cum Laude,** *awarded highest distinction for my PhD*

2018 **DSWeb 2018 Dynamical Systems Software 1st place**
*Award given by the Dynamical Systems division of SIAM for my submission DynamicalSystems.jl (€500)*

## INVITED TALKS

2026 **DynamicsDays2026, Portugal,** *invited to present as a guest speaker in the minisymposium "Response and tipping behaviour in complex systems: from analytic to data-driven methods" by Manuel Santoz Gutierrez*

2026 **Solicited talk, EGU2026,** *for the session AS1.15 "Advancing understanding of the circulation-coupling and Lagrangian evolution of clouds", convened by Matthias Tesche*

2026 **Transient Dynamics: Theory and Applications, Potsdam Institute for Climate Impact Research (PIK)**
*invited by Kalel Luiz Rossi to give an invited talk and a hands-on workshop at the 3-day conference*

2026 **DynamIC, Imperial College London**
*invited by Martin Rasmussen, talk title: "Nonlocal stability and continuation of dynamical systems"*

2025 **SAMOSA consortium (remote)**
*invited by Jacob Haqq Misra, talk title: "Multistability in complex climate datasets"*

2025 **Laboratoire de Météorologie Dynamique (LMD, Paris), France**
*invited by Florent Brient, talk title "Stratocumulus-cumulus transitions in a cloudy energy balance model*

2025 **TUDelft, Geosciences & Remote Sensing department, Netherlands**
*invited by Geet George, talk title: "Stratocumulus-cumulus transitions sensitivity on environmental factors"*

2025 **[Colloquium] University of Oxford, Atmospheric, Oceanic and Planetary Physics Department, UK**
*invited by Milan Klower, talk title: "Stratocumulus-cumulus transitions sensitivity on environmental factors"*

2024 **Max Planck Institute for Meteorology, Germany,** *invited by Hauke Schmidt, talk title: "Stratocumulus-cumulus transitions sensitivity on environmental factors"*

2023 **JuliaHEP (Erlangen Centre for Astroparticle Physics), remote**
*invited by Tamas Gal, talk title: "Reproducibility in science: why it matters and how to achieve it"*

2023 **[Colloquium] University of Utrecht, Netherlands**
*invited by Swinda Falkena, talk title: "Cloud controlling factors and causal timeseries analysis"*

2023 **Max Planck Institute for Evolutionary Biology, Pleon, Germany,** *invited by Maria Ramirez, talk title: "Finding and continuing attractors and their basin fractions for arbitrary dynamical systems"*

2023 **Stockholm University Meteorology Department (MISU), Sweden,** *invited by Frida Bender and Rodrigo Caballero, talk title: "Cloud controlling factors and causal timeseries analysis"*

2022 **Minisymposium "Stability indicators and machine learning", Dynamics Days 2022**
*invited by Nahal Sharafi, talk title: "Stability indicators in DynamicalSystems.jl"*

2022 **Statista (Hamburg division), Germany,** *invited by Jeremiah Lasquety-Reyes, talk title: "Why you should do your agent based modelling with Agents.jl"*

2022 **Rey Juan Carlos University, Spain**
*invited by Alexandre Wagemakers, talk title: "An open approach to nonlinear dynamics"*

2022 **Technical University Munich, weekly seminar of dynamics group (remote)**
*invited by Niklas Boers, talk title: "Nonlinear dynamics software for everyone"*

2022 **University of Oldenburg, Germany**
*invited by Ulrike Feudel, talk title: "An open approach to nonlinear dynamics"*

2021 **Max Planck Institute for Dynamics and Self-Organization, Germany**
*invited by Michael Wilczek, talk title: "Earth's albedo and its symmetry"*

2020 **35th CERES-NASA Science Team Meeting (remote)**
*invited by Normal Loeb, talk title: "Earth's albedo symmetry and cloudiness"*

2019 **Technical University Vienna, Austria**
invited by Stefan Rotter, talk title: "Phase space analysis of quantum transport in graphene"

2019 **University of Nottingham,** *invited by Philip Moriarty, talk title: "Music timeseries analysis: universal structure and its impact on the listening experience"*

2019 **8$^{Th}$ Recurrence Plot Symposium, Zhenjiang, China**
*invited by Norbert Marwan, talk title: "Fresh approach to dynamical systems software"*

2019 **Max Planck Institute for the Physics of Complex Systems, Dresden, Germany**
*invited by Holger Kantz, talk title: "Music timeseries analysis"*

2019 **Potsdam Institute for Climate Impact Research, Germany,** *invited by Norbert Marwan, talk title: "Spatiotemporal timeseries prediction using reconstructed local states"*

2018 **Technical University Munich, Germany**
*Invited by Oliver Junge, talk title: "Fresh approach to dynamical systems software"*

2018 **University of Regensburg, Germany**
*invited by Jonathan Eroms, talk title: "nonlinear resonances in ballistic transport in antidot superlattices"*

## EDITORIAL WORK

2026 - ongoing **Reviewer for the Alexander von Humboldt Foundation,** *for the Division of Physics, Engineering, Mathematics*

2026 **Expert assessor for UKRI's "UK Metascience Unit",** *invited to provide expert assessments on the novelty of selected publications within my area of expertise (nonlinear dynamics)*

2026 – ongoing **Section Editor for "Methods in Climate Science" of the journal PLOS Climate**

2024 – 2026 **Editorial Advisory Board member for CHAOS***, until end of 2026*

2017 - ongoing **Reviewer Work,** *I have reviewed 31 articles for various journals such as Journal of Open Source Software, Journal of Climate, Geophysical Research Letters, CHAOS, PLOS One, Physical Review E, Physical Review Research, Social Network Analysis and Mining, Science Advances, and more*